

Efficacy of CRC Senior Services Report

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2026-01-20

Introduction

I have been tasked with discovering if a certain service is effective in warding off decline in the senior citizens' population group at the CRC. Right now, the data is a sample from the larger effect group. Clients are measured on a scale from 4 to 16, with 4 showing the ability of the client to live independently while a score of 16 strongly suggests the client be in assisted living. Each test consists of 4 parts scored 1-4. A secondary and more simple goal is to show the percentage of clients in which this improves or maintains their scores.

Firstly, I will import the data and perform any data cleaning required for the analysis.

```
library(readxl)

## Warning: package 'readxl' was built under R version 4.4.3

dat_raw <- read_xlsx("C:/Users/charl/Desktop/Work for CRC/ILA data for Stats.xlsx")

## New names:
## * '' -> '...9'
## * '' -> '...10'
## * '' -> '...11'
## * '' -> '...12'

# I want to remove the columns that do not contain any data
dat <- dat_raw[,1:8]

# We do not need the raw set anymore
rm(dat_raw)

diff_scores <- dat$`Score Difference`
```

First Task: Efficacy

We want to determine if this service wards off decline or maintains scores for the clients, within a 5% error, which is the industry standard. We will use a one sided paired t-test to determine this. We want to test against a null hypothesis that the difference of scores from before and after the service is administered is 0, with an alternative hypothesis that states the difference of scores from before and after are greater than 0. Then, we test to accept or reject the null hypothesis. Let μ_1 be the mean of the scores before the service and μ_2 be the mean of the scores after.

$$H_0 : \mu_1 - \mu_2 = 0$$

$$H_a : \mu_1 - \mu_2 > 0$$

Data Assumptions

In order for the test to produce meaningful results, we must have a couple of requirements fulfilled. Let's define the mean of the distribution for the differences as μ_d . μ_d must firstly follow an approximately normal distribution, each pair of data points must be independent (no dependence between clients), and the original raw data's pairs must be correctly paired.

Independence Based on how the data is collected, I can assume that the test administrators made sure that all answers were their own and that there was no help given during the test, both by other clients and by test administrators. Thus, we can assume by virtue of the test design that all data points are independent.

Normality For the paired T-test, we assumed that the distribution that the mean of the differenced data comes from is approximately normal. In our case, we can rely on the central limit theorem to deliver us the confirmation of our assumptions. The central limit theorem states that with independent samples, all with some degree of measurable variability, that as the number of samples increases, the distribution that these samples come from become more and more normal. That being said, since we have independent samples, and we have a standard sufficient amount of samples to be normally distributed ($n = 44$) according to the Central Limit Theorem.

Paired T-test

Now we can implement a one sided paired T-test to see if the service implemented has an effect on our sample.

```
test <- t.test(dat$`Post Test Score`, dat$`Pre Test Score`, paired = T, alternative = "greater")
print(test)
```

```
##
## Paired t-test
##
## data: dat$`Post Test Score` and dat$`Pre Test Score`
## t = 0.56126, df = 43, p-value = 0.2888
## alternative hypothesis: true mean difference is greater than 0
## 95 percent confidence interval:
## -0.181377      Inf
## sample estimates:
## mean difference
##      0.09090909
```

With a p-value of 0.2888, we fail to reject the null hypothesis that there is no difference in scores before and after the service is administered. Thus, there is no evidence to suggest that the service increases the scores of the clients, meaning there is a lack of evidence to suggest that the service allows patients to get worse than they were before the service. Note that the logical jump to “This service works definitively” would be a supposition from the vacuously true statement the data gives in the hypothesis test, and is a weak argument. The data only shows that there is not much evidence that the service allows a decline in patients’ independence. More trials are required to show the true effect of the service on the population at the CRC, but this is a good sign from the data.

Second Task: Percentage

We also want to see what percentage of this sample sees a maintenance of scores or a decrease. This is a pretty simple calculation that I will include below.

```
count <- 0
for(i in 1:nrow(dat)){
  if(dat$`Score Difference`[i] <= 0) {
    count <- count + 1
  }
}
p <- count/nrow(dat)
print(p)
```

```
## [1] 0.8409091
```

```
1-p
```

```
## [1] 0.1590909
```

We can see that the proportion of those whose scores improved or stayed the same is 84.09%. Let's consider what this means; 15.9% of elderly individuals showed that the service to increase or maintain independent living did not work against the 84.1% of individuals in which it did work. Currently there is no way to affirm whether the service works to the extent of which the designers desire, which would require a clinical study of decline in normal populations vs the sample that receives the service. Our previous analysis does not detect a statistically significant change in the population before and after the service is administered.

In conclusion, while more data is required for more definite results, the data provided is consistent with the service's intended goal of preventing decline. In the future, with more data points, other statistical tests can be put in place to confirm or refute this conclusion.